



King AbdulAziz University

Faculty of Computing & Information Technology

Computer Science Department

Final Lab Exam (10-12-2020)

Course: CPCS-202 (Programming 1)

Term: 2021 First Semester

Time: 120 Minutes (06:00-08:00 P.M)

Instructions:

- Students must complete their task and upload the solution **within 120 minutes.**
- **Submission Link will close after 120 minutes**
 - Java class should be named as follows: FinalExamSectionStudentID, Example: (FinalExamBCR123456).
- Upload ONLY the Java file on Blackboard.
- **Make sure to add your name / ID / Section / course name / Final Lab Exam, as comments at the beginning of your program.** It is important and count in your marks.
- During exam keep saving your work.
- In case of any issue contact the instructor immediately.

Objective:

You are going to develop a program for a Perfumes Shop which allows the customer to select from the available perfumes then calculate the cost and print the bill.

Program Details:

- 1- The name of the perfume shop is **your name**, it appears in the top of perfume menu.
- 2- The shop contains 3 perfumes: Bamboo, Si and Gris with prices **200 SAR, 400 SAR and 600 SAR** respectively.
- 3- The customer can select from the following options (**B for Bamboo, S for Si and G for Gris, C to confirm** the order and print the bill).
- 4- Your program should accept **both small and capital** i.e. B or b, S or s, G or g, C or c.
- 5- If the customer enters **incorrect choice**, the following message should be displayed **“Wrong Selection! Try again”** and allow the customer to enter a choice again.
- 6- The menu should be displayed **iteratively** to allow the customer selecting more than one perfume until the customer enters **C**.
- 7- After the customer select the perfume (B, S, or G), the program should ask to enter **Quantity**.
- 8- You should keep the ordered quantity of each perfume. The customer can order the same perfume many times.
- 9- If the customer chooses C to finish the program, then check if he/she already **select perfumes or not**.
 - a. **If all perfumes quantities are zeros**, that means the customer doesn't select anything. So, print the following message: **"Nothing Selected. See you again."** Then end the program.
 - b. **Otherwise:**
 - Ask the customer to enter his/her **name and mobile number**. Mobile number should be **10 digits** and start with **0**. It should be validated by using **"isCorrectMobile"** method. If it is wrong, allow the customer to enter it again until it becomes correct.
 - Also, the program will generate **order number** by using **"generateOrderNum"** method.
 - The bill should display all **perfumes with their quantities and item total**.

Item	Quantity	Item Total
------	----------	------------
 - Then, **Subtotal, Tax, Total Price with "SAR"**

- **Item total** will be calculated as:

Item total = price of selected perfume * total quantity of this perfume

- **Subtotal** equals summation of all items total.

- There is a **Tax** equals to **15%** added to the final total price.

Tax = Subtotal * 0.15

Tax is calculated by using "**calculateTax**" method.

- The **total price** equals Subtotal + Tax
- Print "**Thank you and see you again**" at the end.

10- You should **declare and use** three methods in your program:

Method Name	Return type	Description
isCorrectMobile(String mobileNum)	boolean	Check if mobile number is correct. Return true if it's length is 10 digits and the first digit is 0, otherwise return false.
calculateTax(int subtotal)	double	It receives subtotal and returns 15% of it.
generateOrderNum(String name)	String	It generates and return order number by: Generating two digits random number_First letter of the name Example: If name is "Noor" then order number would be 74_N

-🔔 Output Sample: see next page →

RUN#1: (with wrong enter + nothing chosen)

```
-----  
Welcome to YOUR NAME Perfumes Shop  
-----  
B: Bamboo, GUCCI      200SR  
S: Si, Giorgio Armani  400SR  
G: Gris, Dior          600SR  
C: Confirm and Exit  
-----  
Enter your selection: H  
Wrong Selection! Try again  
  
Enter your selection: a  
Wrong Selection! Try again  
  
Enter your selection: c  
Nothing Selected. See you again.
```

RUN#2:

```
-----
Welcome to YOUR NAME Perfumes Shop
-----
B: Bamboo, GUCCI          200SR
S: Si, Giorgio Armani     400SR
G: Gris, Dior             600SR
C: Confirm and Exit
-----
Enter your selection: H
Wrong Selection! Try again

Enter your selection: B
Quantity:1
-----
Welcome to YOUR NAME Perfumes Shop
-----
B: Bamboo, GUCCI          200SR
S: Si, Giorgio Armani     400SR
G: Gris, Dior             600SR
C: Confirm and Exit
-----
Enter your selection: S
Quantity:1
-----
Welcome to YOUR NAME Perfumes Shop
-----
B: Bamboo, GUCCI          200SR
S: Si, Giorgio Armani     400SR
G: Gris, Dior             600SR
C: Confirm and Exit
-----
Enter your selection: b
Quantity:2
-----
Welcome to YOUR NAME Perfumes Shop
-----
B: Bamboo, GUCCI          200SR
S: Si, Giorgio Armani     400SR
G: Gris, Dior             600SR
C: Confirm and Exit
-----
Enter your selection: C
Enter your Name: Noor
Enter your Mobile Number (must be 10 digits and start with 0): 0555555
Enter your Mobile Number (must be 10 digits and start with 0): 5055555555
Enter your Mobile Number (must be 10 digits and start with 0): 050555555555
Enter your Mobile Number (must be 10 digits): 0555555555
```

See next page to complete of run#2 sample output →



Hi Noor, order number:74_N

-----Details-----

Item	Quantity	Item Total
Bamboo, GUCCI	3	600
Si, Giorgio Armani	1	400
Gris, Dior	0	0

Subtotal: 1000

Tax:150.0

Total Price:1150.0 SAR

Thank you and see you again.